

$= 0.62 \text{ m s}^{-1}$

[A1]

5 (a)

A and B: in phase

[A1]

A and C: in antiphase (accept completely or 180° or π out of phase.) [A1]

5 (b)

Use stroboscope (B1) and adjust flash frequency for slow motion/ expect to see A moving up as C moves down etc.

Or use a video camera and replay in slow motion/ expect to see A moving up as C moves down etc.

5 (c)(i)

The amplitude of the wave is constant for progressive wave [B1] but varies from 0 at the node to $2A$ at the antinode for stationary wave. {B1}

5 (c)(ii)

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Reflection give rise to waves propagating in both directions [B1].

Interference between these progressive waves of the same amplitude, frequency and speed moving in opposite direction gives rise to stationary wave. {B1}

6 (a)